Exploratory Data Analysis (EDA) Project on Student Performance Dataset

1. Introduction:

Exploratory Data Analysis (EDA) is the process of analyzing datasets to summarize their main characteristics using statistical methods and visualizations. This project analyzes student performance data to identify patterns, trends, and relationships.

2. Objectives:

To analyze student performance data and identify trends, correlations, and key influencing factors using visualizations and statistical summaries.

3. Dataset Description:

The dataset contains information about students including:
Attendance, Study Hours, Midterm Score, Final Score,
Sleep Hours, and Grade.

4. Data Preprocessing:

- Checked missing values
- Verified data types
- Cleaned dataset

5. Statistical Summary:

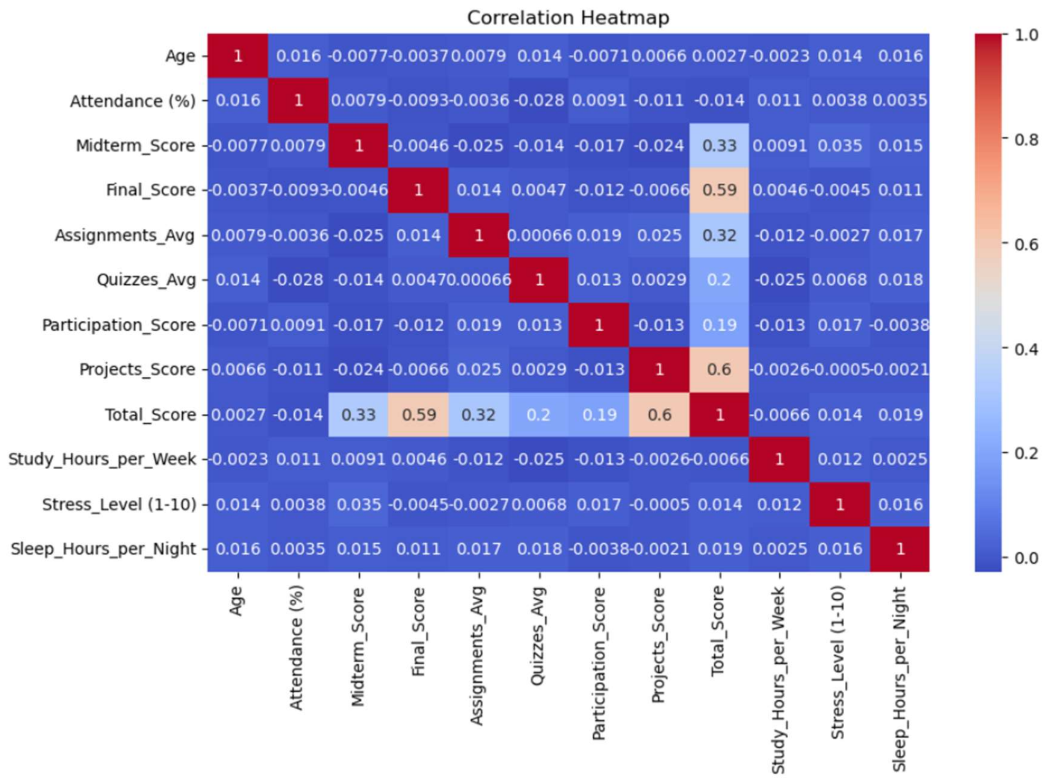
	Age	Attendance (%)	Midterm_Score	Final_Score
count	5000.000000	5000.000000	5000.000000	5000.000000
mean	21.048400	75.356076	70.701924	69.546552
std	1.989786	14.392716	17.436325	17.108996
min	18.000000	50.010000	40.000000	40.010000
25%	19.000000	62.945000	55.707500	54.697500
50%	21.000000	75.670000	70.860000	69.485000
75%	23.000000	87.862500	85.760000	83.922500
max	24.000000	100.000000	99.990000	99.980000

	Assignments_Avg	Quizzes_Avg	Participation_Score	Projects_Score
count	5000.000000	5000.000000	5000.000000	5000.00000
mean	74.956320	74.836214	49.963720	74.78305
std	14.404287	14.423848	28.989785	14.54243
min	50.000000	50.000000	0.000000	50.00000
25%	62.340000	62.357500	25.075000	61.97000
50%	75.090000	74.905000	49.600000	74.54000
75%	87.352500	87.292500	75.500000	87.63000
max	99.990000	99.990000	100.000000	100.00000

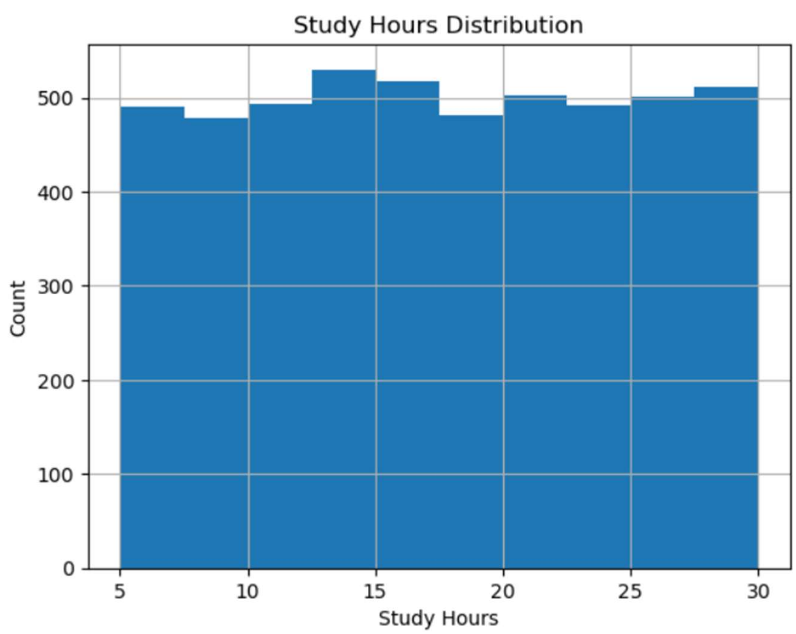
	Total_Score	Study_Hours_per_Week	Stress_Level (1-10)
count	5000.000000	5000.000000	5000.000000
mean	71.652097	17.521140	5.507200
std	7.230097	7.193035	2.886662
min	50.602000	5.000000	1.000000
25%	66.533875	11.500000	3.000000
50%	71.696250	17.400000	6.000000
75%	76.711625	23.700000	8.000000
max	95.091500	30.000000	10.000000

	Sleep_Hours_per_Night
count	5000.000000
mean	6.514420
std	1.446155
min	4.000000
25%	5.300000
50%	6.500000
75%	7.800000
max	9.000000

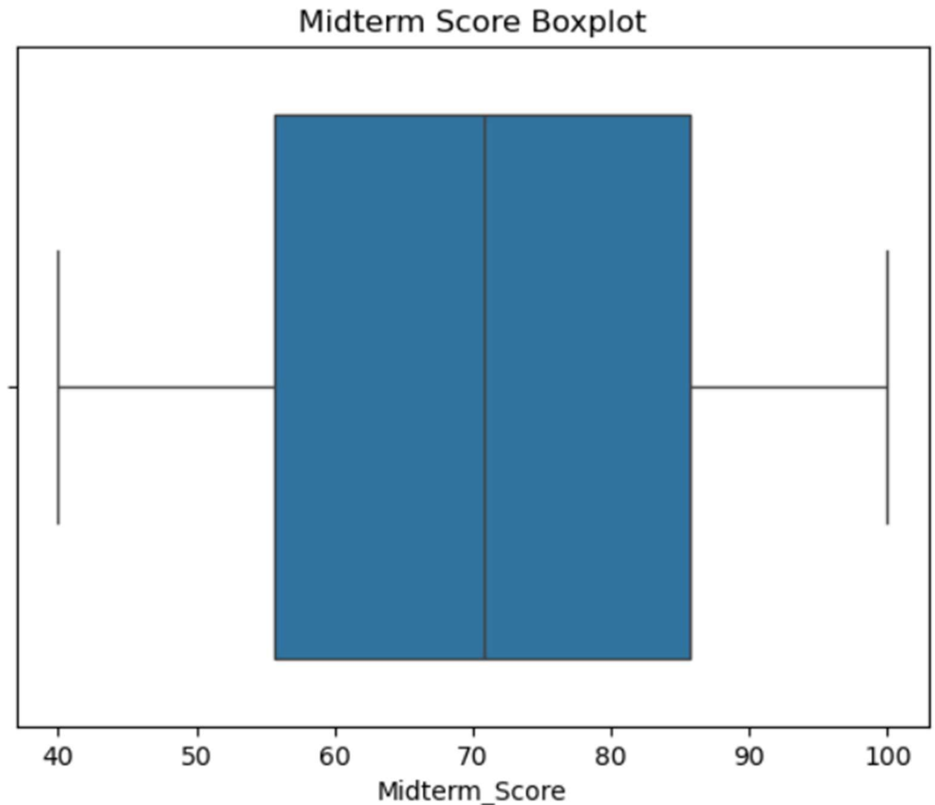
6. Visualizations:



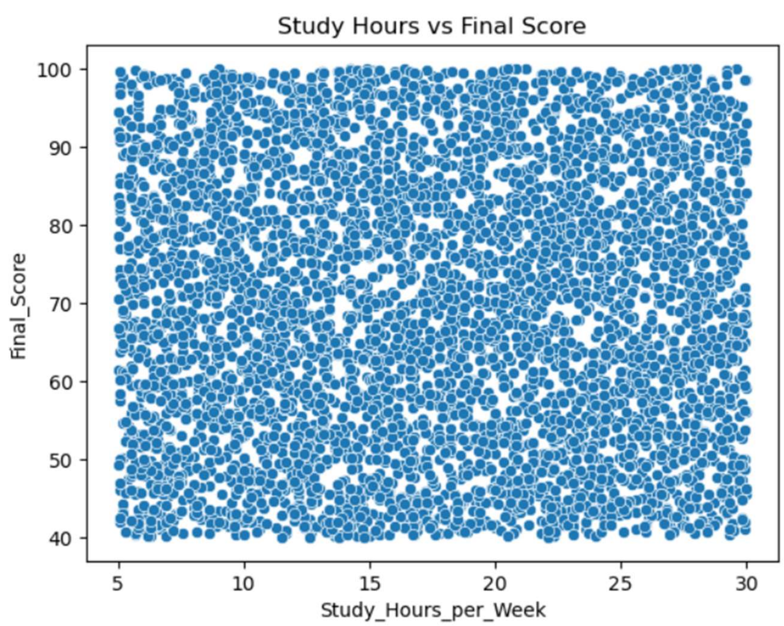
Heatmap



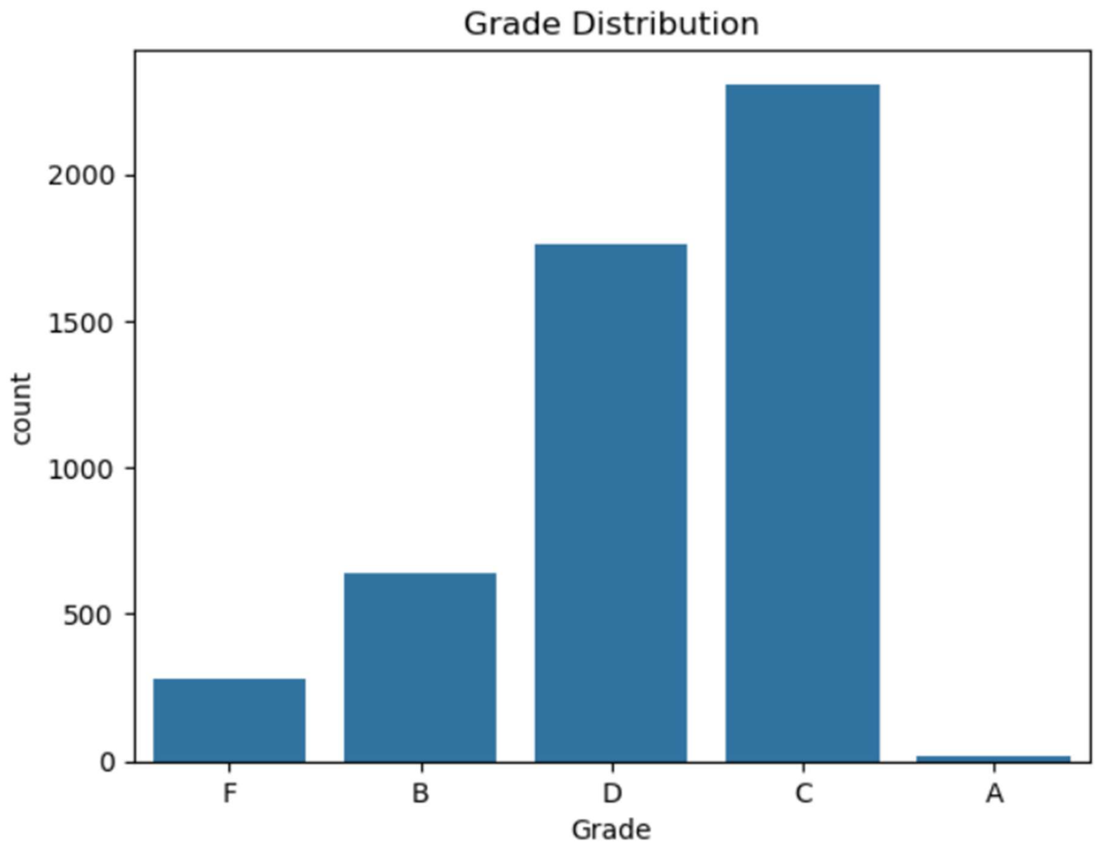
Histogram



Boxplot



Scatterplot



Countplot

7. Insights:

- Students with higher study hours achieved better scores.
- Attendance positively impacts final performance.
- Some outliers were identified in midterm scores.
- Study hours and final score have strong positive correlation.

8. Conclusion:

The EDA process helped identify important trends and relationships in student performance data. Visualizations and statistical analysis provided meaningful insights into factors affecting academic results.